Powder Technology 274 (2015) 431–440

Contents lists available at ScienceDirect

Powder Technology

journal homepage: www.elsevier.com/locate/powtec

Sticking/rebound criterion for collisions of small adhesive particles: Effects of impact parameter and particle size

Sheng Chen, Shuiqing Li *, Mengmeng Yang

Key Laboratory for Thermal Science and Power Engineering of Ministry of Education, Department of Thermal Engineering, Tsinghua University, Beijing 100084, China

ARTICLE INFO

Article history:
Received 14 November 2014
Received in revised form 19 January 2015
Accepted 23 January 2015
Available online 29 January 2015

Keywords:
Discrete element method
Micro-particle
Oblique collision
Sticking velocity
Energy dissipation

ABSTRACT

The collision-sticking phenomenon among micron-sized particles is ubiquitous in nature and plays an important role in formation of particle agglomerates or deposits. In this paper, a validated 3D JKR-based discrete-element method (DEM) is employed to investigate oblique collisions of micron-sized particles. Special attention is paid to the effects of impact parameter and particle size on the sticking/rebound criterion and energy dissipation pathways. Various energy dissipation mechanisms, including viscoelastic effect, sliding resistance and rolling resistance, are incorporated in our DEM model. Based on our simulation results, the temporal evolution for the collision process is revealed in detail to establish a deeper understanding of collision dynamics. Moreover, two regimes are clarified according to the effect of impact angles on the sticking criterion. In the low impact angle regime, the normal critical velocity V_{CN} keeps as a constant. While in the regime of high impact angle, V_{CN} rapidly drops. Furthermore, the strong effect of particle size ratio on the critical velocity is also analyzed. This effect can be well described by the dimensionless adhesion parameter, *Ad*. Finally, a generalized formula is drawn as the sticking/rebound criterion for collisions of fine particles, which can be readily implemented in computational fluid dynamics (CFD) codes.

© 2015 Elsevier B.V. All rights reserved.

1. Introduction

Collision and coagulation of micron particles, existing in a variety of areas of engineering, biology, astrophysics and environmental science, play a central role in determining the structure of particle agglomerates or deposits [1–4]. The question whether or not a collision may result in sticking is of importance for the evolution of agglomerates [5,6]. In the size range of 10 microns or smaller, the van der Waals force between particles is more significant than the gravitational force or fluid force and is primarily responsible for holding the dust particles together [7]. The interaction between colliding particles becomes quite complicated due to the presence of the adhesive force.

In order to describe coagulation processes properly, a physical understanding of binary collisions is prerequisite. During the collision process with adhesion taken into account, relative motions (e.g., normal overlapping, relative sliding, rolling and twisting) between contact particles take place accompanied by diverse energy-loss mechanisms including internal friction, viscoelastic response, interfacial slip, etc [8–11]. Owing to the adhesion force between contact particles, a gentle collision with an impact velocity less than the critical sticking velocity V_C will result in sticking. And both the viscoelastic damping force and the adhesive force contribute to the energy loss [12,13]. On the other hand, colliding particles will bounce off each other when collision velocities exceed V_C. By introducing the coefficient of restitution e, defined as the ratio of the relative post- and pre-collision velocities, the outcome of an energy-dissipated bouncing collision can be quantitatively described.

Various experimental and theoretical studies have been published over the decades on the critical sticking velocity V_C and the coefficient of restitution e [14–17]. Among those experimental studies, one of the most extensive data for normal collisions between micron particles and a surface is from Wall et al. [16]. Güttler et al. provided an overview on published literatures describing experiments on normal collisions, with a focus on the relationship between the coefficient of restitution and the incident velocity [18]. However, due to the small size of micron particles and the ultra-short time scale associated with particle collisions, the inability to observe the detailed information and the failure to account for tangential or rotational velocities during collision experiments have greatly limited the physical understanding of the processes. From the theoretical level, simplified analytical solutions for the critical sticking velocities were proposed by both Dominik and Tielens [2] and Thornton and Ning [17]. However, only the energy required to separate contact surfaces was considered in their simplified models, and the interaction between particles was considered to be elastic, which is inconsistent with the practical situation. In a real collision, many factors, including the viscoelasticity of materials, the collision angles and the

* Corresponding author. Fax: +86 10 62773384.
E-mail address: lishuiqing@tsinghua.edu.cn (S.Q. Li).

http://dx.doi.org/10.1016/j.powtec.2015.01.051
0032-5910/© 2015 Elsevier B.V. All rights reserved.

size of particle, contribute to the sticking criterion. Therefore, it's hard to predict V_C with a single formula.

To better understand these collision problems, it is essential to propose an appropriate micro-dynamic model describing the interactions between contact micron particles. Dynamic models generally describe the impact process through a combination of quasi-static contact theories and dissipation mechanisms [1,7,19]. The theory on quasi-static contact usually serves as a fundamental description of the contact, while energy dissipation mechanisms account for the dynamic effects such as viscoelasticity of materials. A variety of models exist to describe the effect of van der Waals adhesion on the elastic force during static contact of particles. Among these models, the JKR (Johnson, Kendall and Roberts), DMT (Derjaguin, Muller and Toporov) and M-D (Maugis-Dugdale) models are widely accepted and applied for regimes with different Tabor parameter λ_T [20–22]. The JKR model assumes that the adhesive force acts only inside the contact region [20] and is appropriate for relatively large, compliant particles for which the Tabor parameter λ_T is large compared to unity [1]. Based on the JKR model, the dynamic process of normal impacts between micron particles and a flat surface was described by Liu et al. and the results have been validated by classic particle/surface impact experiments [12,16]. Liu et al. also concluded that the JKR model can give proper predictions of contact size and overlap even in conditions beyond the expected JKR range [23]. Recently energy dissipated in head-on collisions between two spheres was further investigated by Krijt et al. [13], in which a collision model with combined energy-loss mechanisms including adhesion, viscoelasticity and plastic deformation was presented.

However, aforementioned studies mostly focused on *normal* collisions between particles or between a particle and a surface. Generally, the collisions between two spheres are oblique with non-zero impact angles which may give rise to a highly coupled inter-particles motions including normal overlapping, sliding and rolling. The notion of normal critical velocity V_{CN} has been exploited as a sticking/rebound criterion without consideration of the effect of rolling and sliding motions in the tangential degree. Thus, this simplistic criterion has a serious limitation for collisions with high impact angle θ. Konstandopoulos found that, when the impact angle is larger than a critical value, the sticking cannot happen despite that $V_N < V_{CN}$ is met [24]. A three-dimensional simulation has also shown that the approach of energy conversion due to tangential motions is quite sensitive to the impact angle and will affect the outcome of a collision [25]. Despite the efforts made to model tangential friction problems caused by oblique collision [10,11,26], an explicit expression of the sticking criterion V_{CN}, which is of great importance for incorporation into CFD codes or simulations describing particle deposition and coagulation processes, is really needed but still not drawn out yet.

More recently, Li and Marshall [27], and Marshall [28] developed a three-dimensional discrete-element method for adhesive micron-sized particles based on the JKR model (termed as adhesive DEM hereinafter), and successfully applied it to dynamic simulations of micro-particle deposition on both flat and cylindrical surfaces with a series of experimental validations [7,29]. By using this adhesive DEM approach, not only macroscopic parameters (e.g., the restitution coefficient and the critical sticking velocity) can be predicted, but also microscopic information (e.g. the contact area, the normal overlapping and the relative sliding distance) during a collision can be provided. This DEM approach enables us to further explore a generalized expression for the sticking/rebound criterion of collisions covering all impact angles.

In addition, changing of particle size will lead to a variation of relative importance of particle inertia and adhesive force. With particles' size increasing from micrometer to millimeter or even larger, a transition from adhesion-dominant regime to inertia-dominant regime will give rise to quite different results of collisions [30]. Fractions of energy dissipated due to various mechanisms also change with the particle size. And this significant size-effect on the outcome of a collision can also be investigated by the adhesive DEM.

In this paper, we focus on oblique collisions between micron particles by employing a three-dimensional adhesive DEM. The central issue addressed in this paper is whether particles of a certain size will stick or not upon collisions with a given impact parameter and how the kinetic energy of colliding particles is dissipated. The structure of the paper is as follows: the brief introduction of adhesive DEM, including the JKR-based models together with various energy dissipation mechanisms, is given in Section 2.1 and Section 2.2; then the collision geometry is given in Section 2.3; and the results and discussion, including microscopic information of oblique collisions and the effects of impact parameter and particle size on the sticking criterion are presented in Section 3.

2. Models and method

2.1. Adhesive DEM

The current investigation involves simulations of inter-particle collisions by employing a three-dimensional DEM approach for adhesive particles. In this adhesive DEM, particles are regarded as soft spheres whose momentum and angular momentum equations are solved by incorporating various models to describe the contact force, for an overview see, e.g. [1,7,27]. In this work, only adhesive contact forces $\mathbf{F}_A$ and torques $\mathbf{M}_A$ during particle collision are taken into account. The JKR model was employed to describe the elastic contact force F_{ne} as a combination of adhesion and elastic deformation of particles. The expressions for F_{ne} and normal overlap δ_N are written in terms of the contact region radius $a(t)$ [1],

$$\frac{F_{ne}}{F_C} = 4\left(\frac{a}{a_0}\right)^3 - 4\left(\frac{a}{a_0}\right)^{3/2}, \quad \frac{\delta_N}{\delta_C} = 6^{1/3}\left[2\left(\frac{a}{a_0}\right)^2 - \frac{4}{3}\left(\frac{a}{a_0}\right)^{1/2}\right], \tag{1}$$

where the critical force and overlap, F_C and δ_C, and the equilibrium radius, a_0 are given by

$$F_C = 3\pi\gamma R, \quad \delta_C = \frac{a_0^2}{2(6)^{1/3}R}, \quad a_0 = \left(\frac{9\pi\gamma R^2}{E}\right)^{1/3} \tag{2}$$

which are all expressed in terms of effective parameters, including the particle radius $R = (r_i^{-1} + r_j^{-1})^{-1}$, the elastic moduli $E = ((1-\nu_i^2)/E_i + (1-\nu_j^2)/E_j)^{-1}$, the shear moduli $G = ((2-\nu_i)/G_i + (2-\nu_j)/G_j)^{-1}$ and the surface energy $\gamma = \sqrt{\gamma_1\gamma_2}$.

Apart from the JKR elastic contact force, we consider the viscoelasticity of materials. The solid-phase dissipation force is assumed to be proportional to the rate of change of the material deformation in the form

$$F_{nd} = \eta_N \mathbf{v}_R \cdot \mathbf{n} = \eta_N \frac{d\delta_N}{dt}. \tag{3}$$

Here, the normal dissipation coefficient $\eta_N = \alpha(m^* k_N)^{1/2}$ is described in literature [9] and $\mathbf{v}_R = (\mathbf{v}_i + \boldsymbol{\Omega}\times\mathbf{r}_i) - (\mathbf{v}_j + \boldsymbol{\Omega}\times\mathbf{r}_j)$ is the relative velocity at the contact point on particle surfaces. For head-on collisions between two bulk particles with negligible adhesion or plastic deformation, the viscoelasticity effect is the only dominant factor to dissipate energy and the restitution coefficient is almost kept in a fixed value, e_0, which is related to the damping coefficient α. Both e_0 and α can intrinsically reflect the damping properties of adhesionless particles, with their relationship studied in literatures [1,28]. However, as for collisions between micron-sized particles in presence of adhesion, both inter-particle adhesion and the damping properties affect the outcome of the collision [12–14]. As a result, the restitution coefficient e ranges from 0 for sticking collisions to e_0 for inertia-dominant collisions. Here, we set the surface energy $\gamma = 15\ \mathrm{mJ/m^2}$ and the coefficient $e_0 = 0.6$ to account for the damping effect. Then a sixth-order formula $\alpha =$

$1.2728 - 4.2783\,e_0 + 11.087\,e_0^2 - 22.348\,e_0^3 + 27.467\,e_0^4 - 18.022\,e_0^5 + 4.8218\,e_0^6$ derived in [28] is used to calculate the corresponding coefficient α.

In almost any collision, sliding occurs as a motion of particles in the direction perpendicular to the axis connecting the particle centers. We use the spring-dashpot-slider model proposed by Cundall and Strack [26] to calculate the sliding resistance

$$F_S = -\min\left[\left|k_T\boldsymbol{\delta}_T\cdot\mathbf{t}_S + \eta_T\mathbf{v}_S\cdot\mathbf{t}_S\right|, \mu_f\left|F_{ne}+2F_C\right|\right]. \tag{4}$$

Here, the first term of sliding resistance due to the spring is proportional to the relative sliding displacement $\boldsymbol{\delta}_T = \int_{t_0}^{t}\mathbf{v}_S(\tau)d\tau$, where $\mathbf{v}_S$ is the sliding velocity given by $\mathbf{v}_S = \mathbf{v}_R - (\mathbf{v}_R\cdot\mathbf{n})\mathbf{n}$. The tangential stiffness coefficient k_T is derived by Mindlin [8] as $k_T = 8Ga(t)$. The dashpot force is linearly proportional to $\mathbf{v}_S$ with an assumption $\eta_T = \eta_N$ [9].

Apart from the sliding resistance, adhesive rolling resistance was also employed using a simple JKR-based model [10], giving

$$M_r = -k_R\xi = -4F_C(a/a_0)^{3/2}\xi, \tag{5}$$

where $\xi = \left(\int_{t_0}^{t}\mathbf{v}_L(\tau)d\tau\right)\cdot\mathbf{t}_R$ is the rolling displacement. For binary collision problems in our study, the expression for the rolling velocity $\mathbf{v}_L$ of particle i is given by [31]

$$\mathbf{v}_L = -R\left(\boldsymbol{\Omega}_i - \boldsymbol{\Omega}_j\right)\times\mathbf{n} - \frac{1}{2}\left(\frac{r_j - r_i}{r_j + r_i}\right)\mathbf{v}_S. \tag{6}$$

Once ξ reaches a critical value ξ_{crit}, the contact region will start moving at the same velocity as the particle centroid, with a constant rolling displacement equal to ξ_{crit}. Consequently, provided $\xi \geq \xi_{crit}$, the rolling resistance is given by

$$M_r = -4F_C(a/a_0)^{3/2}\xi_{crit}. \tag{7}$$

The critical rolling displacement ξ_{crit} is usually expressed in terms of a critical rolling angle $\theta_{crit} = \xi_{crit}/R$, which has a typical value of 0.01 for micron particles according to the AFM experiment [32].

It is noted that all the parameters used in aforementioned models have been validated using experimental data [7,29]. And the 3D DEM has been successfully applied to simulations of various adhesive particle behaviors, including particle–wall collisions [7] and deposition of particles on a fiber [29] et al.

2.2. Energy dissipation mechanisms

Collisions between real particles are always accompanied by diverse energy-loss mechanisms rather than elastic. How fast the initial kinetic energy is dissipated exerts a tremendous influence on the magnitude of the critical sticking velocity V_C or the coefficient of restitution e in bouncing collisions. Therefore, the energy dissipation mechanisms are discussed in the following.

In the normal degree, the kinetic energy of colliding particles is transferred through the relative motion during the contact period, which can be obtained through integrating the force at some displacement

$$\Delta E_N = \Delta E_{ne} + \Delta E_{nd} = \int_{t_0}^{t}\left(F_{ne} + F_{nd}\right)\dot{\delta}_N(\tau)d\tau. \tag{8}$$

Here, ΔE_{ne} is the conservative part of transferred kinetic energy, including both the stored elastic energy and the released surface energy, and will fluctuate as the colliding particles vibrate back and forth towards each other [13]. ΔE_{nd} represents the amount of energy dissipated due to viscoelasticity and will continuously increase during a collision.

The reduction of the kinetic energy due to the sliding resistance is given by

$$\Delta E_S = -\int_{t_0}^{t}\left(F_S\mathbf{t}_S\cdot\mathbf{v}_S(\tau) + r_iF_S(\mathbf{n}\times\mathbf{t}_S)\cdot\boldsymbol{\Omega}(\tau)\right)d\tau. \tag{9}$$

The friction force between two relatively sliding surfaces steadily dissipates the kinetic energy. Meanwhile, the sliding torque $r_iF_S(\mathbf{n}\times\mathbf{t}_S)$ makes the sliding motion and rolling motion highly coupled and causes a reciprocal conversion between the translational energy and the rotational energy of the particle.

The kinetic energy dissipated due to the rolling resistance is calculated via

$$\Delta E_r = -\int_{t_0}^{t}M_r\mathbf{t}_R\cdot\boldsymbol{\Omega}(\tau)d\tau. \tag{10}$$

All of the energy equations are solved in time steps sufficiently small compared with the collision time scale, so that the temporal evolution of energy dissipation can be obtained.

2.3. Collision geometry

We focus on binary collisions of two initially nonrotating particles. Collisions between a particle and a wall, which can be regarded as a sphere of infinite radius and mass, are also contained in this paper. In order to capture all possible configurations at the instant of contact, appropriate parameters are set to describe the collision geometry.

Firstly, a reference frame which is at rest relative to the target particle (particle 2 in Fig. 1a) is selected so that the relative collision velocity is equal to the velocity of the colliding particle (particle 1 in Fig. 1a). Since the relative position of the particles only matters with respect to the direction of their relative velocity $\mathbf{V}$. We introduce the angle θ between velocity of particle 1 and the $\mathbf{n}$ direction as the only parameter that determines the geometry of a collision due to the axial symmetry. Then the dimensionless impact parameter $\hat{b}$ is defined as the distance b between the path of particle 1 and the center of particle 2 divided by the sum of the two particles' radii, which is given by

$$\hat{b} = \frac{b}{r_1 + r_2} = \sin\theta. \tag{11}$$

The impact parameter ranges between 0 and 1, where $\hat{b} = 0$ signifies a head-on collision and $\hat{b}\to 1$ corresponds to the state that particles barely touches each other. For a collision between a particle and a wall (Fig. 1b), the angle θ is defined in the same way and the impact parameter is given by $\hat{b} = \sin\theta$.

At the first moment of contact t_0, the normal overlap δ_N, the sliding displacement δ_T and the rolling displacement ξ are all set to be zero.

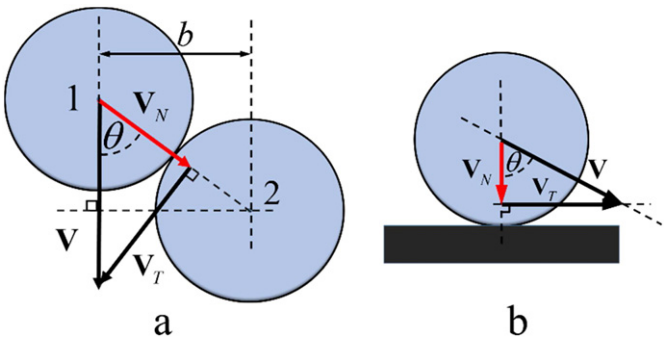


Fig. 1. Illustration of the collision geometry described by the eccentric distance b and the impact angle θ for collisions between (a) two particles and (b) a particle and a wall.

And the initial rate of change for these parameters are set using the impact parameter $\hat{b}$, written as

$$\begin{aligned}\mathbf{v}_R(t_0) &= \mathbf{V}\\ \mathbf{v}_S(t_0) &= \mathbf{V}_T = |\mathbf{V}|\hat{b}\cdot\mathbf{t}_S\\ \mathbf{v}_L(t_0) &= -\frac{1}{2}\left(\frac{r_2-r_1}{r_2+r_1}\right)\mathbf{V}_T = -\frac{1}{2}\left(\frac{r_2-r_1}{r_2+r_1}\right)|\mathbf{V}|\hat{b}\cdot\mathbf{t}_R.\end{aligned} \tag{12}$$

Then the contact forces and torques are calculated using the models list above. The physical parameters used in our DEM simulations are summarized in Table 1. All the variables including displacements, velocities and forces are updated in sufficiently small time steps.

3. Results and discussion

3.1. Temporal evolution during collisions

In DEM simulations of particle collisions, the collision time scale characterizes the response of a particle during the contact period. The collision time scale for an elastic collision between nonadhesive particles is given by $T_{NE} = d_p(\rho_p^2/E_p^2V_N)^{1/5}$, where E_p is the particle Young's modulus [9]. For micron particles in a sticking case, a damped oscillation happens during the contact period due to the presence of inter-particle adhesion [12]. The period of the damped oscillation is regarded as a characteristic time scale for particle motion, and can be obtained by analyzing the governing equation for the normal overlap during the collision between a particle and a wall, which is written as

$$m\frac{d^2\delta_N}{dt^2} + F_{nd} + F_{ne} = 0. \tag{13}$$

When the equilibrium state is nearly achieved, the overlap is almost equal to the final value δ_0 (corresponding to a position of $a = a_0$ and $F_{ne} = 0$). A linear approximation is made to the force and the radius of the contact region a.

$$\begin{aligned}\delta_N &= \delta_0 + \Delta\delta\\ a &= a_0 + \Delta a\\ F_{ne} &= F_{ne}(\delta_0) + \Delta F_{ne} = \Delta F_{ne}.\end{aligned} \tag{14}$$

Take these expressions into the JKR model in Eq. (1), then the linear dependence of the deviation value Δa and ΔF_{ne} on $\Delta\delta$ is obtained after the Taylor expansion is performed. The motion Eq. (13) reduces to a typical damped oscillation form,

$$\frac{d^2\delta_N}{dt^2} + \alpha\left(\frac{4Ea_0}{3m}\right)^{1/2}\cdot\frac{d\delta_N}{dt} + \frac{F_C}{m\delta_C}\cdot\delta_N = 0. \tag{15}$$

The oscillation period is then given by

$$T_N = \frac{2\pi}{\sqrt{F_C/m\delta_C - \alpha^2 Ea_0/3m}}. \tag{16}$$

Scaling analysis gives $F_C/m\delta_C \gg \alpha^2 Ea_0/3m$, thus Eq. (16) is reduced to $T_N \approx 6.72\gamma^{-1/6}R^{7/6}\rho_p^{1/2}E^{-1/3}$. Substituting the variables in Table 1 for a 1 μm particle, we get the result $T_N = 0.13$ μs as the collision time scale. Compared with the time scale for a Hertz collision, our newly derived expression is independent of the impact velocity and has a relation to the surface energy γ. The reason is that, the initial kinetic energy of the colliding particle is mainly dissipated during the first one or two oscillation cycles, and then the particle motion is controlled by the balance between the van der Waals adhesion and the elastic repulsion. Due to the special nature of the van der Waals adhesion, Eq. (16) is more suitable to characterize the time scale associated with the motion of an adhesive particle during a collision.

Table 1
Parameters used in DEM simulations.

Physical parameter	Value	Units
Particle radius (r_p)	1 or larger	μm
Particle density (ρ_p)	2500	kg/m^3
Surface energy (γ)	15	mJ/m^2
Particle Young's modulus	2×10^8	Pa
Restitution coefficient (e_0)	0.6	1

In order to illustrate how the adhesive DEM can generate microscopic information needed to improve the understanding of collision problems, simple cases of oblique collisions between a 1 μm particle and a wall (Fig. 1b) are investigated before applying our models to binary collisions. Through these cases, the temporal evolutions of the displacements in different degrees of freedom are compared, and the relationship between sliding and rolling motions together with the energy dissipation pathways are shown. Here, two typical cases with the impact angle $\theta = 45°$ are carried out. The impact velocities are set as $V_N = V_T = 0.8$ m/s for the sticking case (Fig. 2a, b, c) and $V_N = V_T = 1.0$ m/s for the rebound case (Fig. 3a, b, c).

In the case of lower impact velocity, the kinetic energy is dissipated during oscillating till the final captured state is achieved. The evolution of displacements is shown in Fig. 2a, where the time and the displacements have been normalized by the oscillation time scale T_N given in Eq. (16) and the particle radius r_P respectively. Due to the dissipation of the kinetic energy, in the rebound phase, the particle is not able to overcome the critical overlap $-\delta_C$ (indicated by the horizontal dash line in Fig. 2a and Fig. 3a), at which point the neck connecting the contact surfaces will suddenly break. Then the particle oscillates back and forth towards the final equilibrium point $\delta_N = \delta_0$. As for the sliding motion, the displacement increases rapidly and then reaches an equilibrium value at approximately $T^* = 0.62$. The rolling motion lasts a much longer time compared with the sliding, and the final state is achieved at about $T^* = 3.5$. To further understand the relationship between sliding and rolling motions, we calculate the evolution of the tangential velocity V_T and the rotating velocity, which is defined as the product of particle's rotating rate Ω and particle radius r_P (Fig. 2b). According to the relation between these two velocities, the whole period can be divided into three stages. At the first stage labeled A, V_T is obviously larger than $\Omega\cdot r_P$, which indicates that the surface of the particle slips relative to that of the wall. In this stage, the sliding resistance serves as a more significant mechanism to dissipate the kinetic energy in contrast to the rolling resistance. This sliding force acts at the contact point in a direction opposite to V_T hence gives rise to the decrease of V_T and the increase of $\Omega\cdot r_P$. Then the collision process goes into a transition stage (labeled B in Fig. 2b), during which V_T is very close, but still not identical to $\Omega\cdot r_P$. At this stage, both sliding and rolling resistance work together to dissipate the particle's kinetic energy. During the final stage labeled C, the two lines coincide with each other, which indicates a so-called *pure rolling* motion. The rolling resistance becomes the main mechanism to dissipate the kinetic energy until the final state is achieved.

The evolution of various energies in this collision system, normalized by the initial kinetic energy of the particle, is shown in Fig. 2c. The initial kinetic energy is steadily dissipated through the normal damping force, the sliding resistance and the rolling resistance, which is shown as the continuously increase of ΔE_{nd}, ΔE_S and ΔE_r in Fig. 2c. The conservative part of transferred kinetic energy ΔE_{ne}, which is the sum of released surface energy and stored elastic energy, fluctuates as the colliding particles vibrate back and forth towards each other till the final state is achieved.

As for the impact with higher velocity, shown in Fig. 3a, b, and c, the pre-collision kinetic energy is larger than that could be dissipated during contact, the particle overcomes the critical overlap $-\delta_C$ with a nonzero rebound velocity which results in a final separation at $T^* = 0.73$. Due to the short duration of contact, the evolutions of V_T and $\Omega\cdot r_P$ are

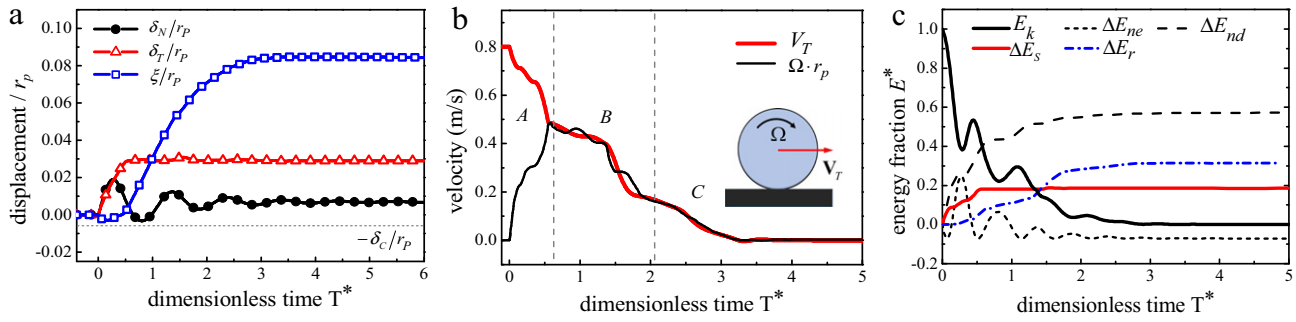


Fig. 2. Plots of the temporal evolution of (a) the normal overlap δ_N, the sliding displacement δ_T and the rolling displacement ξ, (b) the tangential velocity V_T and the rotating velocity $\Omega \cdot r_p$, (c) energy conversion pathways during a typical sticking collision with velocities $V_N = V_T = 0.8$ m/s.

limited to the first stage (Fig. 3b), during which the sliding motion is the dominated dissipation mechanism rather than the rolling. The collision thus can be somehow regarded as a scrape between the contact surfaces. Furthermore, unlike the sticking case, a sudden conversion of energy from ΔE_{ne} to ΔE_{nd} appears at the separation point in Fig. 3c as a result of the sudden disappearance of the necking between contact surfaces. During this sudden change, the energy stored in the contact region is instantaneously released leading to the so called first-contact-loss.

3.2. Prediction of the critical velocity

Firstly, binary collisions between two 1 μm particles, with physical parameters list in Table 1, are investigated. Fig. 4 shows the critical sticking velocity, V_C, increases as the angle θ increases (black line). Previous studies stated that the rolling and sliding motions played only a minor role in collisions between two particles implying a fixed value for the normal component of the critical velocity V_{CN} [2]. However, from the results of the detailed DEM simulation, this conclusion is valid only in the case of small impact angles and breaks down when the impact angle is larger than a certain value. The red line in Fig. 4 gives the variation of V_{CN} with θ. In the regime of small θ, V_{CN} almost holds as a constant and then starts to dramatically drop since the point $\theta \approx 65°$. A characteristic angle θ_{90}, at which point the normal critical velocity $V_{CN}(\theta_{90})$ is equal to 90% of the value at $\theta = 0°$, is defined as a typical angle for the necessity of considering tangential effects on the value of V_{CN}. In this special case, θ_{90} is approximately equal to 72.7° (corresponding to an impact parameter of $\hat{b}_{90} = 0.955$).

Under the conditions of small θ, the normal dissipation is the dominant mechanism that determines the criterion for sticking. Thus we can study this problem with good accuracy by restricting the analysis to the vertical degree of freedom only. When the energy in the vertical degree of freedom is smaller than that could be dissipated by the normal damping force, the particles will stick. Thornton and Ning [17] assumed the interaction between particles to be elastic and used the JKR model to calculate the energy needed to separate the surfaces. As the red line in Fig. 5 shows, the loading path covers only $\delta_N \geq 0$ (curve AB), while the unloading process (curve BAC) can continue till the critical separation point $\delta_N = -\delta_C$. Such loading/unloading process results in the so-called first-contact energy loss, given by

$$\Delta E_{ad} = \int_A^C F_{JKR} d\delta_N = 7.09\left(\frac{w^5R^4}{E^2}\right)^{1/3}, \tag{17}$$

where $w = 2\gamma$ is the work of adhesion. Then a sticking/rebound criterion is obtained as

$$V_{C1} = \left(\frac{14.18}{m_*}\right)^{1/2}\left(\frac{w^5R^4}{E^2}\right)^{1/6}. \tag{18}$$

For two particles $r_1 = r_2 = 1$ μm Eq. (18) leads to $V_{C1} = 0.34$ m/s, which is much lower than the DEM simulation result because of the neglect of the viscoelastic effect. In fact, the overall energy loss ΔE_k consists of two parts, the first-contact energy loss ΔE_{ad} and the energy loss due to the viscoelastic dissipation ΔE_{diss}. ΔE_k can be numerically calculated as the production of the force and the displacement given by

$$\Delta E_k = \int_{-\delta_C}^{\delta_{max}} \Delta F d\delta_N. \tag{19}$$

When the viscoelastic effect is considered, the loading/unloading process is described by the curve A_1BA_2C in Fig. 5, and ΔF in Eq. (19) is the difference between the force at a certain δ_N in loading process and that at the same δ_N in unloading process. To better clarify the energy

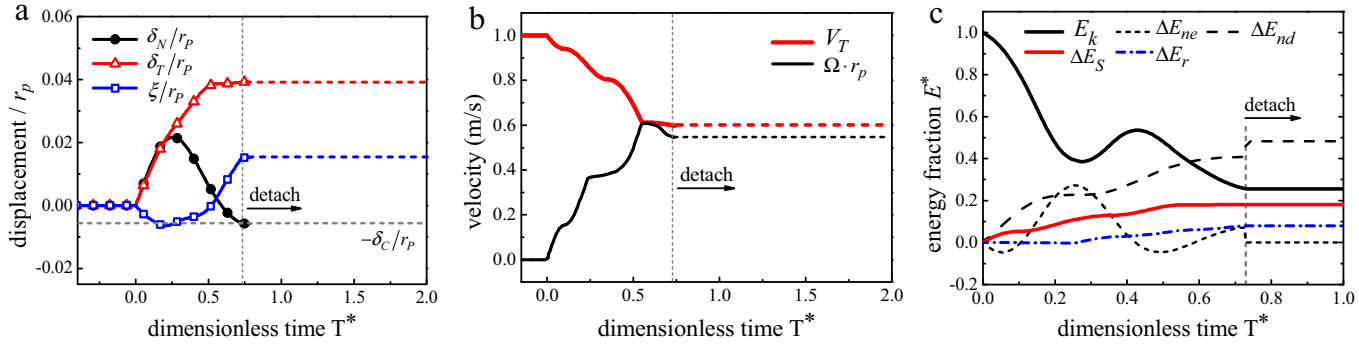


Fig. 3. Plots of the temporal evolution of (a) the normal overlap δ_N, the sliding displacement δ_T and the rolling displacement ξ, (b) the tangential velocity V_T and the rotating velocity $\Omega \cdot r_p$ (c) energy conversion pathways during a typical rebound collision with velocities $V_N = V_T = 1$m/s, the vertical dash line indicates the separation point.

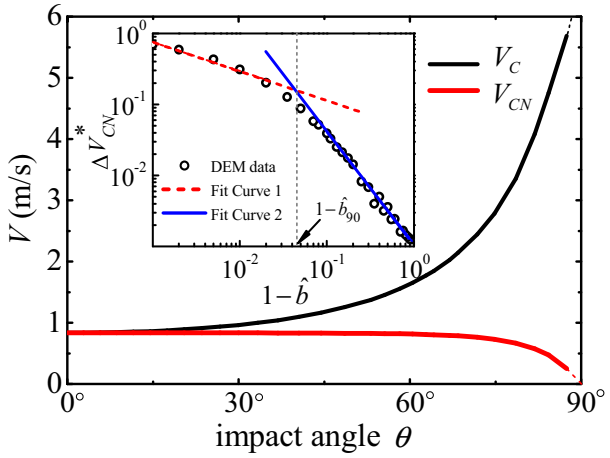


Fig. 4. The critical sticking velocity V_C and its normal component V_{CN} as functions of the impact angle θ.

dissipation mechanism, Fig. 6 shows the relation between normalized ΔF and δ_N, then the energy loss ΔE_k is the sum of the area enclosed by the solid black line and the x-axis. The adhesion part ΔE_{ad}, which is indicated by the area under the JKR line, was used by Thorton and Ning to calculate the critical sticking velocity [17]. However, as shown in Fig. 6, ΔE_{ad} is much smaller than the total energy loss ΔE_k indicating that the viscoelastic effect cannot be neglected. Unfortunately, the dependence of ΔE_{diss} on V is quite complicated due to the complex $F \sim \delta_N$ relationship, thus using a simple formula to predict the criterion seems to be impossible. Reasonable approximations should be taken to calculate ΔE_{diss} for quantitatively clarification of the viscoelastic effect. Focus on a the critical sticking case, the conservation of energy gives

$$\frac{1}{2}m*V_{C0}^2 = \Delta E_k = \Delta E_{ad} + \Delta E_{diss} = \frac{1}{2}m*V_{C1}^2 + \Delta E_{diss}, \tag{20}$$

where V_{C0} is the critical sticking velocity for the head-on collision. The viscoelastic dissipation part ΔE_{diss} as indicated by the shadow region is calculated as

$$\Delta E_{diss} = \int_{-\delta_C}^{0} \left(\Delta F(\delta_N) - F_{JKR}(\delta_N)\right) d\delta_N + \int_0^{\delta_{max}} \Delta F(\delta_N)\, d\delta_N. \tag{21}$$

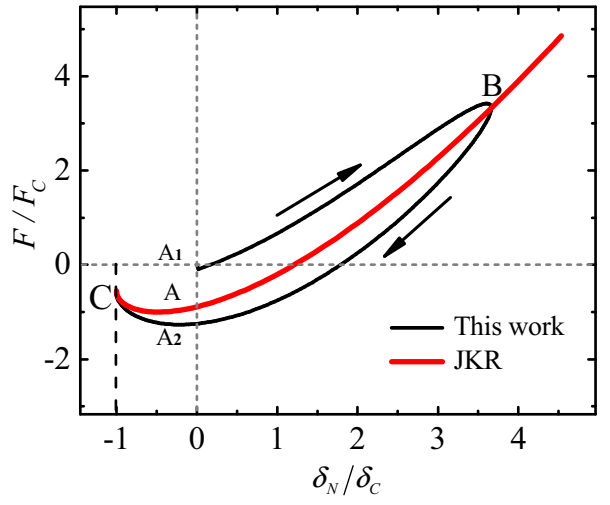


Fig. 5. Evolution of the normalized contact force F/F_C versus the normalized overlap δ_N/δ_C with viscoelastic effect taken into account. Prediction of the JKR model is also plotted (red line).

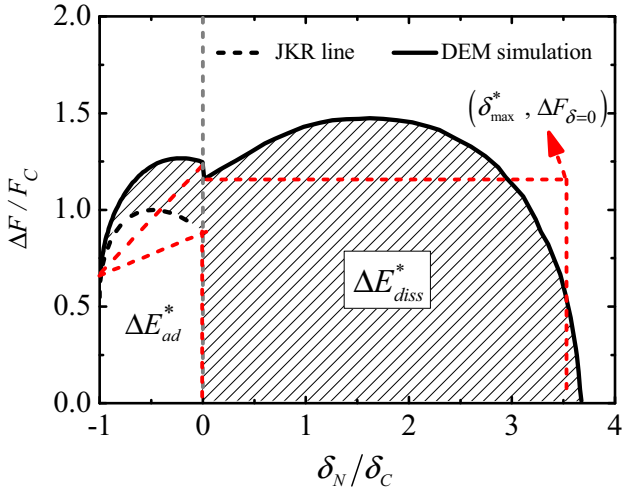


Fig. 6. Relation between normalized ΔF and δ_N. The shadow region represents the normalized viscoelastic energy loss ΔE^*_{diss}, which is approximated using the area enclosed by red dash lines.

As a preliminary approximation, we use $\Delta F(0)$ as a typical value for ΔF during the collision process, then the shadow area can be approximated using the area enclosed by the red dash lines in Fig. 6 (triangle for $\delta_N < 0$ and rectangle for $\delta_N > 0$). We use the Hertzian relation to calculate δ_{max} which gives $\delta_{max} = \left[15m*V_{C0}^2 \Big/ \left(16\sqrt{R}E\right)\right]^{2/5}$. Eq. (21) thus can be expressed in an analytical form

$$\Delta E_{diss} \approx \left(\Delta F(0_-) - F_{JKR}(0)\right)\cdot \delta_C/2 + \Delta F(0_+)\cdot \delta_{max}. \tag{22}$$

Recalling that the damping force is proportional to the particle velocity and ΔE_{diss} is written as

$$\begin{aligned}\Delta E_{diss} &= \Delta E_{diss}(\delta_N<0) + \Delta E_{diss}(\delta_N>0)\\ &\approx \alpha(m*k_N)^{1/2}V_{C1}\cdot\delta_C/2 + \alpha(m*k_N)^{1/2}(V_{C0}+V_{C2})\cdot\delta_{max}\end{aligned}, \tag{23}$$

where $V_{C2} = \sqrt{2\Delta E_{diss}(\delta_N<0) + V_{C1}^2}$ is the critical value for particle velocity at the point $\delta_N = 0$ during the rebound phase, beyond which a final separation will happen. Substituting Eq. (23) into Eq. (20) gives

$$\frac{1}{2}m*\cdot V_{C0}^2 = \frac{1}{2}m*\cdot V_{C2}^2 + \alpha(m*\cdot k_N)^{1/2}(V_{C0}+V_{C2})\cdot\delta_{max}. \tag{24}$$

Rewriting Eq. (24), a simple equation is obtained to calculate V_{C0}, which is written as

$$V_{C0} - V_{C2} = HV_{C0}^{4/5}. \tag{25}$$

Here, V_{C2} serves as a fundamental part of V_{C0}, which reflects the adhesion interaction between contact particles, while the difference between V_{C0} and V_{C2} reflects the viscoelastic effect on the critical velocity and is strongly dependent on the colliding velocities (with a 4/5 power of V_{C0}). The coefficient $H = 3.43\alpha m*^{-1/10}R^{2/15}E^{-1/15}\gamma^{1/6}$ is only determined by the physical parameters of colliding particles.

A graphical method was employed to solve Eq. (25). Replacing V_{C0} with V_{in}, which is regarded as a variable, then the variation of $V_{in} - V_{C2}$ and $HV_{in}^{4/5}$ can be expressed as functions of V_{in} as shown in Fig. 7. The abscissa value of the intersection of these two lines is the solution to Eq. (25) i.e. the value of the critical sticking velocity V_{C0}. The solid red curve represents the case using the same parameters as DEM simulation conducted above, and the corresponding abscissa value at the intersection point is $V_{in} = 0.75$ m/s. Compared with the result $V_{C1} =$

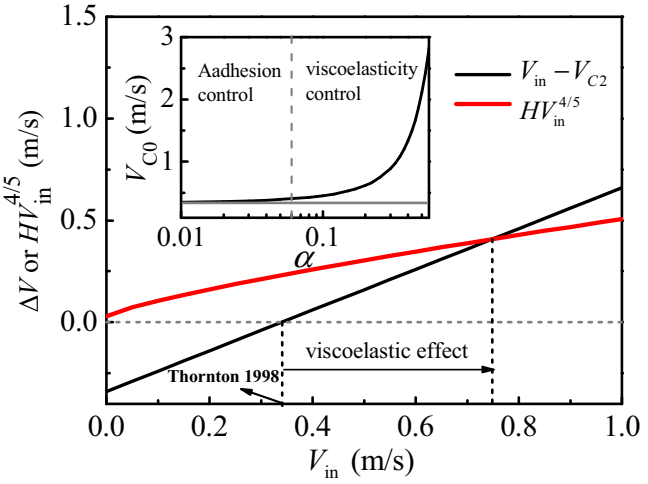


Fig. 7. Plot of $V_{in} - V_{C2}$ and $HV_{in}^{4/5}$ as functions of the incident velocity V_{in}. The abscissa value of the intersection of these two lines is the solution to Eq. (25). The inset depicts the variation of V_{C0} as a function of the dissipation coefficient α.

0.34 m/s calculated using Eq. (18), the result obtained from Eq. (25) is much closer to the numerical calculation $V_{C0} = 0.83$ m/s.

Looking into the expression of the coefficient H, it is found that the dissipation coefficient α has a much stronger influence on H than the other parameters. In order to obtain a bottom-up understanding of the effect of α, a dimensionless form for the collision equation, Eq. (13) of two particles is given as [28]

$$\frac{d^2\hat{\delta}}{d\hat{t}^2} + 2\left(2^{1/4}\right)\alpha\hat{a}^{1/2}\frac{d\hat{\delta}}{d\hat{t}} + 4\left(2^{1/2}\right)\hat{a}^3\left[1-(\hat{a}_0/\hat{a})^{3/2}\right] = 0,$$
$$\hat{\delta} = 2\hat{a}^2\left[1-\frac{2}{3}(\hat{a}_0/\hat{a})^{3/2}\right], \tag{26}$$

where $\hat{a} = a/r_P$ and $\hat{a}_0 = a_0/r_P = (9\pi\gamma/8RE)^{1/3}$. The only parameters in the equation for $\hat{\delta}$ and $\hat{a}$ are the dissipation coefficient α and the dimensionless equilibrium radius $\hat{a}_0$. Therefore the critical sticking velocity V_{C0} for head-on collisions can be regarded as a function of these two parameters giving $V_{C0} = V_{C0}(\hat{a}_0, \alpha)$. The dimensionless equilibrium radius $\hat{a}_0$ is related to the ratio of attractive van der Waals force to elastic rebound force and is equal to 0.091 in this case. Varying the dissipation coefficient α with a fixed $\hat{a}_0$, the variation of V_{C0} is depicted by the black line in the inset of Fig. 7. As α increases, which means the viscoelastic effect getting stronger, the V_{C0} increases monotonically. According to the difference between V_{C0} and V_{C1} (gray horizontal line in the inset of Fig. 7), we define two distinct regimes: (i) For $\alpha < 0.064$, the relative difference $\Delta V^* = (V_{C0} - V_{C1})/V_{C1}$ is less than 20% indicating that the kinetic energy is mainly consumed to overcome the adhesion between surfaces in contact. For materials with α in this adhesion-controlled regime, Thornton's equation is a reasonable approximation to calculate V_{C0}. (ii) For materials with a high α value, where the dissipation due to viscoelasticity effect cannot be neglected, a more accurate prediction of V_{C0} thus should be carried out by solving Eq. (25).

The aforementioned theoretical analysis is valid only for collisions under small impact angle. When θ is larger than the characteristic value of θ_{90} as mentioned above, the energy loss due to tangential motions will directly affect the outcome of a collision thus cannot be neglected. Therefore it is necessary to study in more detail the energy dissipation pathways under different impact angles to show its effect. The energy distributions after collisions between two 1 μm particles with different impact velocities are shown in Fig. 8, where E_K, ΔE_n and ΔE_S represents the remaining kinetic energy, the normal dissipation and the sliding dissipation respectively. No rolling motion happens during collisions between two identical particles according to the conservation of angular momentum. All these energies have been normalized by the initial kinetic energy. For a fixed impact angle θ, the normalized energy losses in sticking cases remain constant even if the impact velocity changes. The reason is that, during these low-velocity collisions, the energy need to be dissipated for sticking is less than that can be dissipated by various dissipation mechanisms. The energy distribution is thus determined by kinetic laws, i.e. the conservations of momentum and angular momentum. When the impact velocity is larger than the critical value for sticking, the interactions between surfaces in contact can no longer dissipate enough kinetic energy to achieve a final sticking. The colliding particles thus will rebound from each other. The fraction of remaining kinetic energy E_K increases monotonously with the impact velocity for these rebound cases due to the limited dissipation abilities. Comparing energy distributions of collisions with different impact angles, we find that the fraction of sliding dissipation ΔE_S increases with θ while the normal part ΔE_n decreases. The relative importance of these two dissipation mechanisms, which can be characterized by the ratio $\kappa = \Delta E_S/\Delta E_n$, is sensitive to the impact angle and significantly affects the outcome of a collision. Fig. 9 shows the relation between the normal critical velocity V_{CN} and the ratio at the critical point $\kappa(V_C)$ for 1 μm particles with friction coefficient μ_f ranging from 0.1 to 0.3. In cases of $\kappa < 1$, which indicates a normal dissipation dominated regime, V_{CN} keeps consistent with the value of a head-on collision and is not influenced by the friction coefficient. However, V_{CN} gradually drops as κ increases further into the regime of sliding-loss controlled, where the curves with different μ_f deviate from each other, since the sliding resistance F_S is strongly dependent on μ_f.

Due to the highly nonlinear nature of oblique collision processes, it is almost impossible to derive a generalized expression for V_{CN} in the high θ regime from theoretical level. Fortunately, simple power laws for variation of V_{CN} can be obtained by fitting the DEM results in Fig. 4 after a simple mathematical transformation. We redraw the DEM data of

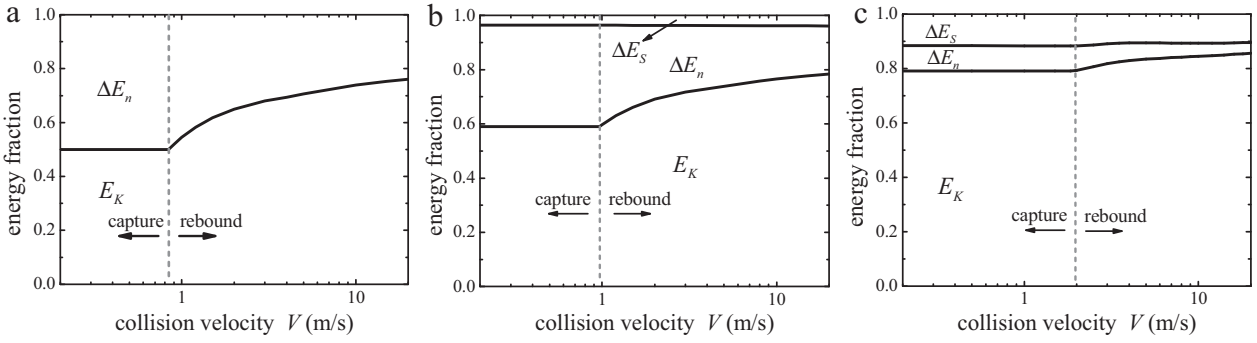


Fig. 8. The fraction of remaining kinetic energy E_K, the normal dissipation ΔE_n and the sliding dissipation ΔE_S after collisions between two 1 μm particles with different velocities under impact angles (a) $\theta = 0°$ $\left(\hat{b} = 0\right)$, (b) $\theta = 30°$ $\left(\hat{b} = 0.5\right)$, and (c) $\theta = 64.15°$ $\left(\hat{b} = 0.9\right)$. The vertical dash line indicates the critical sticking velocities.

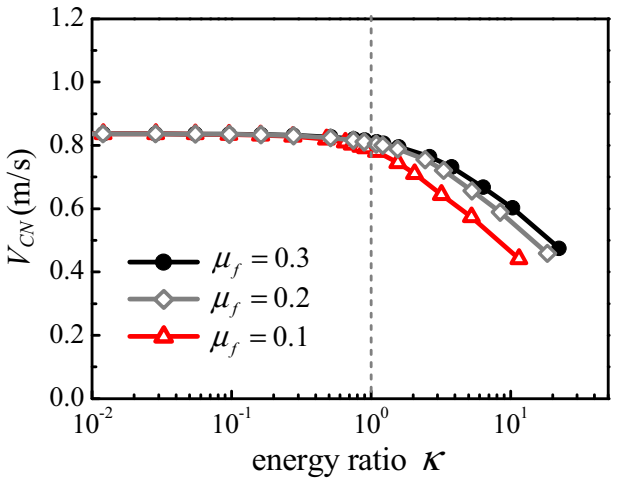


Fig. 9. The relation between the normal component of the critical velocity V_{CN} and the energy ratio κ. The vertical dash line indicates the separation of $\kappa = 1$.

$V_{CN}(\theta)$ in the form of a relative deviation from the value of the head-on (i.e. $\theta = 0$) case V_{C0}, which is written as $\Delta V_{CN}^{*} = (V_{C0} - V_{CN})/V_{C0}$, as a function of $1-\hat{b}$ in double logarithmic coordinates (as shown in the inset of Fig. 4). Two different linear curves are employed to fit the relation between ΔV_{CN}^{*} and $1-\hat{b}$ in two regimes respectively. Then an explicit expression for normal critical velocity V_{CN} under an impact parameter $\hat{b}$ can be obtained,

$$f\left(\hat{b}\right) \equiv \frac{V_{CN}\left(\hat{b}\right)}{V_{C0}} = \begin{cases} 1-0.0011\left(1-\hat{b}\right)^{-1.6} & \text{for } 0 \le \hat{b} \le \hat{b}_{90} \\ 1-0.0454\left(1-\hat{b}\right)^{-0.4} & \text{for } \hat{b}_{90} < \hat{b} \le 1 \end{cases}, \tag{27}$$

where V_{C0} is the critical velocity for head-on collisions and $\hat{b}_{90} = 0.955$ is a dividing point of the two regimes. In the regime of $\hat{b} \le \hat{b}_{90}$, the prefactor is much smaller than unity corresponding to the fact that V_{CN} varies quite slowly in this normal dissipation dominant regime. When the impact parameter further increases into the regime of $\hat{b}_{90} < \hat{b} \le 1$, the tangential effects makes the V_{CN} drop much faster than before. By applying Eq. (27) combined with Eq. (25) determining V_{C0}, we can easily get the normal critical velocity V_{CN} under any given impact parameter $\hat{b}$.

3.3. Effect of size ratio

Particle size is one of the most important parameters for characterizing the behavior of particle systems. Most particle systems cover a wide range of sizes, which emphasizes the need for an appreciation of how collision phenomenon varies with the particle size. Therefore, binary collisions between particles of different sizes are investigated using adhesive DEM assuming that the surface energy γ, elastic moduli E and the dissipation coefficient α are material parameters thus not affected by the particle size. The radii of the particles are recorded as r_1 for the colliding particle and r_2 for the target one respectively. Attention was paid to the particles with size ranging from 1 to 50 μm, where adhesive force plays a key role during collision process. The size ratio β, which is defined as $\beta = r_2/r_1$, is assumed to be larger than unity without loss of generality.

Firstly, we fix $\beta = 1$, which indicates $r_1 = r_2 = r_P$, and find that the normal critical velocities V_{CN} under three representative impact parameters decrease rapidly as r_P increases (as shown in Fig. 10a). For a better explanation, a dimensionless adhesion parameter *Ad*, proposed first by Li and Marshall [1,27], is employed to quantify the size effect. It is defined as the ratio between inter-particle adhesion $2\gamma \cdot \pi R^2$ and particle inertia $1/2m^{*}V^2$. For a general collision with particles' radii of $r_1 = r_P$ and $r_2 = \beta r_P$, the expression for *Ad* can be written as

$$Ad = \frac{2\gamma \cdot \pi R^2}{1/2mV^2} = \frac{3\gamma}{\rho V^2 r_P} \cdot \frac{1+\beta^3}{(1+\beta^2)\cdot\beta}. \tag{28}$$

The variation of *Ad* is also plotted in Fig. 10a as a function of the radius r_P. The varying tendencies of the V_{CN} curves under $\hat{b} = 0$, $\hat{b} = 0.5$ and $\hat{b} = 0.9$ are found to be consistent with each other and can be described well by the *Ad* line throughout this size range. Thus the normal critical velocities for collisions between two identical particles nearly scale like r_P^{-1}.

On the other hand, it is interesting to look at the effect of the size ratio β on the collision results. To study this, the radius of particle 1 is fixed as $r_1 = 1$ μm, and β changes from 1 to 50. The results are shown in Fig. 10b, unlike the monotone decreasing trend shown in Fig. 10a, V_{CN} decreases firstly as β increases until a transition point $\beta \approx 1.7$ is reached, after which V_{CN} continuously increases. The three curves in Fig. 10b show a good consistency with the line of *Ad* in low β region, whereas slightly deviate from each other as β gets further larger. The reason for this deviation is that during an oblique collision between two particles with a large size ratio, the relative rolling motion is triggered to dissipate energy, which makes the collision process less predictable.

With both the size effect and the effect of impact parameter taken into account, an analytical expression for the normal component of the critical sticking velocity $V_{CN}\left(\hat{b}, r_P, \beta\right)$ is given as,

$$V_{CN} = f\left(\hat{b}\right) \cdot \phi(r_P, \beta) \cdot V_{C0} = f\left(\hat{b}\right) \cdot \frac{r_{P0}}{r_P} \cdot \frac{1+\beta^3}{(1+\beta^2)\cdot\beta} \cdot V_{C0}, \tag{29}$$

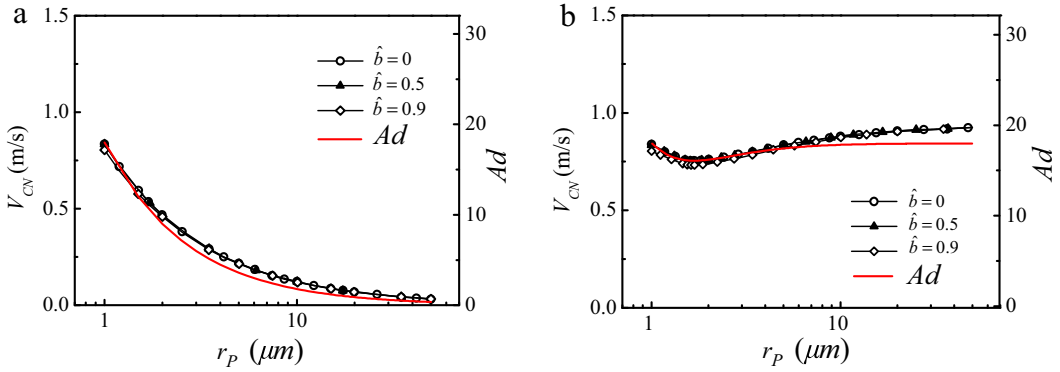


Fig. 10. (a) Plots of the normal critical velocity V_{CN} for (a) collisions with identical particle radius $r_1 = r_2 = r_P$ as a function of r_P, and for (b) collisions with a fixed radius $r_1 = 1$ μm of the smaller particle as a function of the size ratio β. The variation of the adhesion parameter *Ad* is also plotted.

where r_P is the radius of the smaller particle and β is the size ratio. V_{C0} is the critical velocity for a head-on collision between two particles with identical radius r_{P0} and can be obtained by the method described in Section 3.2. Look into Eq. (29), $f\left(\hat{b}\right)$ accounts for the effect of impact parameter and is expressed in Eq. (27) and $\phi(r_P, \beta)$ describes the influence of particle size and size ratio on V_{CN}.

Summarizing the above results, the procedure to calculate the normal critical sticking velocity V_{CN} for collisions under any given impact parameter $\hat{b}$ is as follows:

(1) Use Eq. (18) to calculate the critical velocity V_{C1} due to first-contact energy loss.
(2) Determine the critical velocity at the point $\delta_N = 0$ by $V_{C2} = \sqrt{\alpha(m * k_N)^{1/2} V_{C1}\delta_C + V_{C1}^2}$ with the elastic stiffness k_N given by $k_N = 4Ea_0/3$ with $a_0 = (9\pi\gamma R^2/E)^{1/3}$. After inputting an intrinsic restitution coefficient e_0, the dissipation coefficient α can be determined by a six-order polynomial $\alpha = 1.2728 - 4.2783\, e_0 + 11.087\, e_0^2 - 22.348\, e_0^3 + 27.467\, e_0^4 - 18.022\, e_0^5 + 4.8218\, e_0^6$ as described in [28].
(3) Calculate the coefficient $H = 3.43\alpha m^{*\,-1/10} R^{2/15} E^{-1/15} \gamma^{1/6}$, and then solve Eq. (25) $V_{C0} - V_{C2} = HV_{C0}^{4/5}$ to determine the critical velocity V_{C0} for head-on collisions.
(4) Use $V_{CN}\left(\hat{b}\right) = f\left(\hat{b}\right) \cdot V_{C0}$ with $f\left(\hat{b}\right)$ given in Eq. (27) to calculate the normal critical velocity for collisions under any given impact parameter $\hat{b}$.
(5) Use $V_{CN} = f\left(\hat{b}\right) \cdot \phi(r_P, \beta) \cdot V_{C0}$ with $\phi(r_P, \beta)$ given in Eq. (29) to account for the effect of particle size ratio.

The scheme described above can be easily incorporated into any CFD codes or programs describing particle deposition and coagulation processes to determine the sticking criterion of a general collision between two adhesive micro-particles.

4. Conclusion

In this paper, a three-dimensional DEM approach, incorporating JKR adhesive contact model and the related energy dissipations of viscoelasticity, sliding and rolling resistances, is used to study the oblique collisions between particles at micrometer scale. The effects of impact parameter and particle size ratio on the critical sticking velocity and the energy dissipation pathways are analyzed. The main results are drawn as follows:

(1) DEM results show, the micro-dynamics of an oblique collision between micron-sized particles is revealed as three stages, the sliding-controlled stage, the rolling-controlled one and the between. For sticking cases, the dissipation mechanisms, including viscoelastic effect, sliding and rolling resistances, all exert their contributions to the dissipation of initial kinetic energy. On the other hand, for cases of particle rebound, the rolling resistance can be neglected due to the sufficiently short duration of contact.
(2) The effect of impact angle θ on the normal critical sticking velocity V_{CN} is examined, and two distinct regimes are clarified. In the regime of $\theta < \theta_{90}$, V_{CN} almost keeps as a constant. While as θ further increases into the regime of $\theta > \theta_{90}$, V_{CN} rapidly drops. For collisions in low θ regime, a new simplified analytical method by considering both adhesion and viscoelastic effect is proposed to predict V_{CN}. This method agrees well with the DEM simulated result and shows better accuracy compared with the previous theoretical formula in literatures. In high θ regime, the sliding resistance serves as an significant mechanism for energy dissipation and the decreasing trend of V_{CN} is affected by the friction coefficient μ_f. An explicit expression $f\left(\hat{b}\right)$ is drawn to relate V_{CN} for collisions under any given impact parameter to that of a head-on collision, V_{C0}.
(3) Apart from the size dependence of V_{CN} for mono-sized particles, the role of the size ratio β of binary particle collisions is further clarified. Different from the monotonically inverse correlation between V_{CN} and r_P in mono-sized cases, V_{CN} decreases firstly and then increases as β increases. It can be attributed to the competition between inter-particle adhesion and particle inertia, which is well described by the adhesion parameter, Ad. Finally, by considering the effects of both size ratio and impact parameter, an explicit formula in the form of $V_{CN} = f\left(\hat{b}\right) \cdot \phi(r_P, \beta) \cdot V_{C0}$ is put forward as a sticking/rebound criterion for collisions of micron-sized particles covering all impact angles.

Acknowledgment

This work has been funded by the National Natural Science Funds of China (No. 51390491 and 50976058) and by the National Key Basic Research and Development Program (No. 2013CB228506). We acknowledge Dr. Colin Thornton at Birmingham and Dr. Charley Wu at Surrey for the communication during the 2011 DEM symposium. Thanks are due to Prof. Qiang YAO for helpful discussions. SQL is particularly grateful to Prof. Jeff Marshall for close collaborations on DEM for small adhesive particles.

References

[1] J.S. Marshall, S.Q. Li, Adhesive Particle Flow: A Discrete-Element Approach, Cambridge University Press, 2014.
[2] C. Dominik, A.G.G.M. Tielens, The physics of dust coagulation and the structure of dust aggregates in space, Astrophys. J. 480 (1997) 647–673.
[3] A.G. Konstandopoulos, Deposit growth dynamics: particle sticking and scattering phenomena, Powder Technol. 109 (2000) 262–277.
[4] G.I. Bell, Models for the specific adhesion of cells to cells, Science 200 (1978) 618–627.
[5] F. Brauer, C.P. Dullemond, T. Henning, Coagulation, fragmentation and radial motion of solid particles in protoplanetary disks, Astron. Astrophys. 480 (2008) 859–877.
[6] D. Paszun, C. Dominik, Collisional evolution of dust aggregates. From compaction to catastrophic destruction, Astron. Astrophys. 507 (2009) 1023–1040.
[7] S.Q. Li, J.S. Marshall, G.Q. Liu, Q. Yao, Adhesive particulate flow: the discrete-element method and its application in energy and environmental engineering, Prog. Energy Combust. Sci. 37 (2011) 633–668.
[8] R.D. Mindlin, Compliance of elastic bodies in contact, J. Appl. Mech. 16 (1949) 259–268.
[9] Y. Tsuji, T. Tanaka, T. Ishida, Lagrangian numerical simulation of plug flow of cohesionless particles in a horizontal pipe, Powder Technol. 71 (1992) 239–250.
[10] C. Dominik, A.G.G.M. Tielens, Resistance to rolling in the adhesive contact of two elastic spheres, Philos. Mag. A 92 (1995) 783–803.
[11] C. Dominik, A.G.G.M. Tielens, Resistance to sliding on atomic scales in the adhesive contact of two elastic spheres, Philos. Mag. A 73 (1996) 1279–1302.
[12] G.Q. Liu, S.Q. Li, Q. Yao, A JKR-based dynamic model for the impact of micro-particle with a flat surface, Powder Technol. 207 (2011) 215–223.
[13] S. Krijt, C. Guettler, D. Heisselmann, C. Dominik, A.G.G.M. Tielens, Energy dissipation in head-on collisions of spheres, J. Phys. D. Appl. Phys. 46 (2013) 435303.
[14] P.F. Dunn, R.M. Brach, M.J. Caylor, Experiments on the low-velocity impact of microspheres with planar surfaces, Aerosol Sci. Technol. 23 (1995) 80–95.
[15] O.V. Kim, P.F. Dunn, Direct visualization and model validation of microsphere impact and surface capture, J. Aerosol Sci. 39 (2008) 373–375.
[16] S. Wall, W. John, H.C. Wang, S.L. Goren, Measurements of kinetic energy loss for particles impacting surfaces, Aerosol Sci. Technol. 12 (1990) 926–946.
[17] C. Thornton, Z. Ning, A theoretical model for the stick/bounce behaviour of adhesive, elastic–plastic spheres, Powder Technol. 99 (1998) 154–162.
[18] C. Güttler, D. Heißelmann, J. Blum, Normal collisions of spheres:a literature survey on available experiments, arXiv:1204.0001, 2012.
[19] N.V. Brilliantov, F. Spahn, J.M. Hertzsch, T. Pöschel, Model for collisions in granular gases, Phys. Rev. E 53 (1996) 5382–5392.
[20] K.L. Johnson, K. Kendall, A.D. Roberts, Surface energy and the contact of elastic solids, Proc. R. Soc. Lond. A 324 (1971) 301–313.
[21] K.L. Johnson, J.A. Greenwood, An adhesion map for the contact of elastic spheres, J. Colloid Interface Sci. 192 (1997) 326–333.
[22] B.V. Derjaguin, V.M. Muller, Y.P. Toporov, Effect of contact deformations on the adhesion of particles, Prog. Surf. Sci. 45 (1994) 131–143.
[23] G.Q. Liu, S.Q. Li, Q. Yao, On the applicability of different adhesion models in adhesive particulate flows, Front. Energy Power Eng. Chin. 4 (2010) 280–286.

[24] A.G. Konstandopoulos, Particle sticking/rebound criteria at oblique impact, J. Aerosol Sci. 37 (2006) 292–305.
[25] W. Cheng, R.M. Brach, P.F. Dunn, Three-dimensional modeling of microsphere contact/impact with smooth, flat surfaces, Aerosol Sci. Technol. 36 (2002) 1045–1060.
[26] P.A. Cundall, O.D.L. Strack, A discrete numerical model for granular assembles, Geotechnique 1 (1979) 47–65.
[27] S.Q. Li, J.S. Marshall, Discrete element simulation of micro-particle deposition on a cylindrical fiber in an array, J. Aerosol Sci. 38 (2007) 1031–1046.
[28] J.S. Marshall, Discrete-element modeling of particulate aerosol flows, J. Comput. Phys. 228 (2009) 1541–1561.
[29] M.M. Yang, S.Q. Li, Q. Yao, Mechanistic studies of initial deposition of fine adhesive particles on a fiber using discrete-element methods, Powder Technol. 248 (2013) 44–53.
[30] D. Antypov, J.A. Elliott, B.C. Hancock, Effect of particle size on energy dissipation in viscoelastic granular collisions, Phys. Rev. E 84 (2011) 021303.
[31] K. Bagi, M.R. Kuhn, A definition of particle rolling in a granular assembly in terms of particle translations and rotations, J. Appl. Mech. 71 (2004) 493–501.
[32] B. Sümer, M. Sitti, Rolling and spinning friction characterization of fine particles using lateral force microscopy based contact pushing, J. Adhes. Sci. Technol. 22 (2008) 481–506.